

Don't Panic ... just do it in Parallel!
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Summary

Parallel coordinates is a methodology for visualizing N-dimensional geometry and multivariate problems. In this self-contained up-to-date overview the aim is to clarify salient points causing difficulties, and point out more sophisticated applications and uses in statistics which are marked by **. Starting from the definition of the parallel-axes multidimensional coordinate system, where a point in Euclidean N-space R^N is represented by a polygonal line, it is found that a *point* $\leftrightarrow$ *line* duality is induced in the Euclidean plane R^2 . This leads to the development in the *projective*, P^2 , rather than the Euclidean plane. Pointers on how to minimize the technical complications and avoid errors are provided. The representation (i.e. visualization) of 1-dimensional objects is obtained from the *envelope* of the polygonal lines representing the points on their points. On the plane R^2 there is a inflection-point $\leftrightarrow$ cusp, conics $\leftrightarrow$ conics and other potentially useful dualities. A line $\ell \subset R^N$ is represented by $N - 1$ points with a pair of indices in $[1, 2, \dots, N]$. This representation also enables the visualization and computation of prox-

imity properties like the minimum distance between pairs of lines [18]. The representation of objects of dimension ≥ 2 is obtained *recursively*. Specifically, the representation of a p -flat, a plane of dimension $2 \leq p \leq N-1$ in R^N is obtained from the $(p-1)$ -flats it contains, and which are obtained from the $(p-2)$ -flats and so on all the way down from the points (0-dimensional); hence the recursion. A p -flat is represented by p -points each with $(p+1)$ indices. This is the key message: **** high-dimensional objects may be visualized recursively, in terms of their higher dimensional components, rather than directly from their points.** Further, this process is robust so that “near” p -flats are also detected in the same way and very useful tight error bounds are available. The representation of a smooth hypersurface in R^N is obtained as the *envelope* of the tangent hyperplanes. The set of points obtained in this way visually reveal properties like convexity, whether the surface is developable, or ruled. A simpler but ambiguous representation for hypersurfaces is also given together with modeling applications of an algorithm for computing and displaying interior, exterior or surface points.

Keywords: Multidimensional Visualization, Parallel Coordinates, Visual Data Exploration, Nonlinear Models, Blessings for Dimensionality

”Let no one ignorant of Geometry enter” At entrance of Plato’s Academy

1 Dimensionality is not a Curse.

1.1 The Why, When and Who

The fascination with dimensionality may predate Aristotle and Ptolemy who argued that space has only three dimensions. It was not until the nineteenth century when Riemann, Lobachevsky, Bolyai and Gauss unshackled the imagination with their non-Euclidean and higher-dimensional geometries. The desire to augment our perception, limited as it is by the experience of our three-dimensional habitation, has attracted considerable attention and several visualization methodologies have been developed. There are beautiful non-technical reviews by Tufte in [29], [30] and [31] and together with the bibliographies in [35] and [17], they are indicative of the interest in the field. Visualization provides *insight through images* [23], and can be considered as a collection of application specific mappings:

$$Problem\ Domain \longrightarrow Visual\ Range.$$

The goal is to construct a **visual model** to aid our intuition and our imagination so as to better understand a class of problems. Understanding the

underlying geometry of multidimensional problems can provide crucial insights into what is possible and what is not. For example, in 1917 the physicist Ehrenfest [7] showed that planetary orbits are stable only in dimension 3. Another intriguing dimensionality result is that rotating bodies have an axis of rotation only in odd-integer dimensions ([27] p. 189). For statisticians, the geometrical visualization of multivariate datasets is valuable in understanding and describing their structure.

Professor S.S.Cairns, while giving a course at the University of Illinois in 1959, challenged his students to find a way to visualize higher dimensional geometries. In the spirit of Descartes, I considered constructing a “multidimensional” coordinate system, but how? In geometry the fundamental concept is *parallelism*, which does not require a notion of angle, rather than orthogonality. This and since orthogonality “uses-up” the plane very fast led me to **Parallel Coordinates**. After deriving the elementary *point* $\leftrightarrow$ *line* duality, and in spite of the Professor’s encouragement¹, I did not pursue the idea much until 1977. Then while giving a course on Linear Algebra I was, in turn, challenged by the students to “*show*” higher dimensional spaces for $N \geq 4$; parallel coordinates won the day and the systematic development began [12]², [13], [16], [17], [18], [14]. The goals of the program were and still are the visualization of multivariate *relations* without loss of information together with some other properties that are brought out in the narrative.

**** Parallel coordinates** (abbreviated ||-coords) transform multivariate relations into 2-D patterns, a property that is well suited for visual data exploration and analysis, and are included in several software tools starting with EDA (Chomut [5]), and followed by Finsterwalder[9], VisuLab(Hinterberger[25]), ExplorN (Bolorfoush (88), Carr (92)), WinViZ (Eickemeyer), VisDB(Keim [20]), Xmdv(Ward[21],[33]), XGobi (Swayne, Cook, Buja, [26]), Strata(||-coords by Gleason), Diamond(Rogowitz et al [24]), PVE (Inselberg [1], Adams, Hurwitz, Chatterjee, Austel), Influence Explorer (Tweedie, Spence [32]) and others which include ||-coords. Notably, a major project by Eurostat, the European Union’s Statistical Office, is underway to develop customized data exploration software based on ||-coords. This type of application hinges on :

- an informative display of the data,
- good choice of queries, and
- skillful *interaction* by the user of the display in search of patterns corresponding to relationships among the variables in the data.

Though ||-coords has been successful in data analysis [1], it is not recommended when the variables are mostly categorical. Rather than elaborate

¹Legend has it that young people are most likely to do something when they are *forbidden* from doing so ... rather than when they encouraged.

²Since then Parallel Coordinates has occasionally enjoyed rediscovery by other “inventors” ... confirming the old adage that “success has many parents while failure is an orphan”.

on this, which has already been covered in [15], we will provide pointers and motivation for new applications in this field which take greater advantage of the methodology's strengths. The newest and most powerful implementation of parallel coordinates *ParallaX* (Avidan T. & Avidan S.) is presented in another paper in this issue. Other applications areas including Collision Avoidance Algorithms for Air Traffic Control, Computer Vision, Models for Process Control and Decision Support will be pointed out in due course.

Spurred by the interest in visualization and information visualization in particular, courses including $||$ -coords are regularly taught in various universities and research is blossoming with more and more M.Sc. and Ph.D. theses being under way. It is time then for a didactic overview to demystify the issues which experience has shown to cause grief and misunderstanding.

1.2 The what

This is a good place to describe the fundamental terminology and the process, but please do not let the formalism intimidate you. A relation between N real variables $x_1, x_2, \dots, x_N$ is a subset of R^N – the Euclidean N -Dimensional Space i.e. $F \subset R^N$. In order not to lose information F should be mapped *uniquely* into a planar figure, a subset of R^2 (i.e. a relation between x_1, x_2). Formally, the plan is to construct a mapping

$$\mathcal{J} : 2^{P^N} \rightarrow 2^{P^2} \times \text{Index Set}, \quad (1)$$

where $2^A = \{B \mid B \subset A\}$ is the *power set* of A . The reason for using P^N , the *Projective* N -Dimensional Projective Space, rather than Euclidean N -space as well as the role of the *Index Set* will be explained in due course. It is important to note that *subsets* of P^N are mapped into subsets of P^2 rather than *points into points*. A subset F of P^N is *represented* by $\overline{F} = \mathcal{J}(F)$. We require $\mathcal{J}$ to be *one – to – one* i.e.

$$\mathcal{J}(F_1) = \mathcal{J}(F_2) \Leftrightarrow F_1 = F_2. \quad (2)$$

It is easy to see that in principle it is possible for $\mathcal{J}$ to be 1-1 since the cardinality of both the domain and range is $\aleph_2$. The mapping is constructed *recursively* on the dimensionality of the object F being represented. That is, first for points (0-dimensional), then lines (1-dimensional), followed by p -flats (p -dimensional planes $0 \leq p \leq N - 1$), and then certain hypersurfaces. This mapping is not a *projection*, which in general lose a great deal of information.

In the Euclidean plane R^2 with xy -Cartesian coordinates, N copies of the real line labeled $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_N$ are placed equidistant³ and perpendicular to the x -axis. They are the axes of the Parallel Coordinate system for R^N all having the same positive orientation as the y -axis. The implementation of

³For some applications, we will see later that, different inter-axes distances are more advantageous.

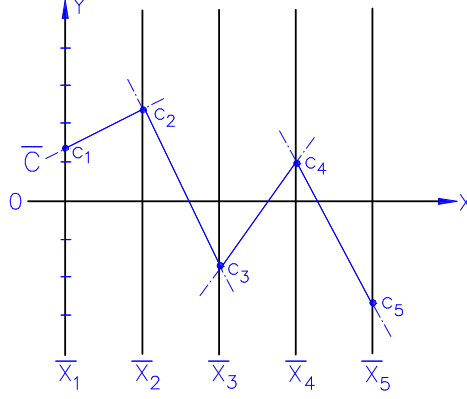


Figure 1: The polygonal line $\tilde{C}$ represents the point $C = (c_1, c_2, c_3, c_4, c_5)$.

the mapping $\mathcal{J}$ will be derived with this *multidimensional coordinate system*. To start with, a point C with coordinates $(c_1, c_2, \dots, c_N)$ is represented by the complete polygonal line $\tilde{C}$ (i.e. the lines of which only the local segments are usually shown) whose N vertices are at $(i-1, c_i)$ on the $\tilde{X}_i$ -axis for $i = 1, \dots, N$ as shown in Fig. 1. In this way, a 1-1 correspondence between points in R^N and planar polygonal lines with vertices on the parallel axes is established. In principle, a large number of (even infinitely many) axes can be placed and be *seen* parallel to each other. The representation of points is deceptively simple. Deeper additional ideas are needed for the visualization of multivariate relations generating a whole **methodology** whose highlights we will cover here. So next time you need to deal with a multivariate problem don't panic, and for heavens sake don't curse dimensionality⁴. Just do it in Parallel!

2 The Plane, Envelopes and Duality

We start by representing 1-dimensional relations (i.e. subsets), typically having infinitely many points, by the *envelope* [2] of the corresponding infinite family of polygonal lines representing the points in the relation. The use of envelopes is the first and most common cause of grief and misunderstandings, but don't despair. Let's look at the situation in 2-D. Points on the x_1x_2 -plane are represented in parallel coordinates by a line on the xy -plane, as shown in Fig. 2. If points are represented by lines it is natural to ask how might lines

⁴It has been reported that in a rare Ecumenical manifestation, the Chief Rabbinate of Israel and the Vatican are preparing a non-denominational blessing for Dimensionality.

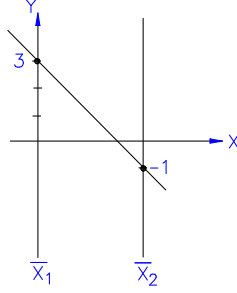


Figure 2: Points on the plane are represented by lines. The line shown represents the point (3,-1).

be represented? We do the experiment shown on the top part of Fig. 3 where points on a line are sampled and plotted as lines. Holy harmonica, all the lines seem to intersect. Is that always true? A line say,

$$l : x_2 = mx_1 + b, \quad (3)$$

is a set of points A which are represented by the set of lines $\bar{A}$ in parallel coordinates (abbreviated by ||-coords). It really turns out that they always intersect⁵ at the point:

$$\bar{l} : \left(\frac{d}{1-m}, \frac{b}{1-m} \right), m \neq 1. \quad (4)$$

where the distance between the parallel axes is d . This point represents the linear *relation*, eq. (3), and is the envelope of the family of lines $\bar{A}$. The point, *all by itself*, suffices to represent the line, for the two parameters m and b specify completely both l and $\bar{l}$. So ||-coords in 2-D induce a *point* $\leftrightarrow$ *line* duality (i.e. mapping points into lines and vice versa) which is illustrated in Fig. 3. But there is a little problem when $m = 1$. Despite appearances, the point $\bar{l}$ does not “blow up” as $m \rightarrow 1$. Rather, when the limiting process is done carefully, one finds that $\bar{l} \rightarrow \infty$ but in the specific direction whose slope is b/d . So lines with $m = 1$ are *not mapped into a point* but into a *direction* and still all the information is there! The fact that it is a direction tells us that $m = 1$ while the slope of the direction gives us the value of b . What is going on? Well, dualities properly reside in the *projective* [6] and not the Euclidean plane. The “directions” are in fact “points” (called “ideal”) of the *projective plane*. One does not need expertise in projective geometry to do ||-coords but awareness is advisable to avoid blunders. Here then is the first instance showing why the mapping $\mathcal{J}$ is taken over the projective space.

⁵So far there are 17 1/2 known ways of proving of this, but rush to contribute yours since the grand prize for the most elegant proof has not yet been awarded.

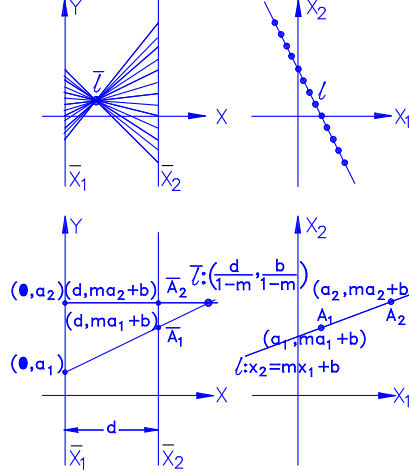


Figure 3: In the plane parallel coordinates induce a point $\longleftrightarrow$ line duality.

Expressing eq. 4, in homogeneous coordinates with triples within [...] and triples within (...) denoting line and point coordinates respectively

$$l : [m, -1, b] \longrightarrow \bar{l} : (d, b, 1 - m), \quad (5)$$

we see that lines with $m = 1$ are represented by the ideal points $(d, b, 0)$. The horizontal position of $\bar{l}$ depends only on the slope m , see eq. 4, so that for vertical lines l (i.e. $m = \infty$) $\bar{l}$ is on the y-axis, while for horizontal lines l (i.e. $m = 0$), $\bar{l}$ is on the x-axis. Lines with negative slope are represented by points within the parallel axes, lines with $0 < m < 1$ with points to the right of the $\bar{X}_2$ -axis and lines with $1 < m$ with points to the left of the $\bar{X}_1$ -axis. Among other things, this points out the reason for using the *whole* line $\bar{P}$ in the representation of P rather than just the segment between the parallel axes, for the point $\bar{l}$ may lie outside the strip between the axes. The dependence of the horizontal position of $\bar{l}$ only on m shows that parallel lines are represented by points on the same *vertical* line, a property which provides some useful applications. There is also a nice construction for detecting orthogonality [13].

The role of envelopes is seen clearly in finding the image(representation) $\bar{r}$ of a curve r . The image is obtained as the *envelope* of the lines representing the points of r . We distinguish between the two by referring to the original curves as *point-curves* and their images as *line-curves*. In Fig. 4 and its *dual* Fig. 5, we see two of the 6 ways that conics(point-curves) map into conics(line-curves). Actually, this is a special case of a more general result involving generalized conics [19] (abbreviated gconics) which are sections of a double cone whose bases are two copies of a bounded convex set. For brevity, referring to bounded convex sets as *estars*, unbounded convex sets

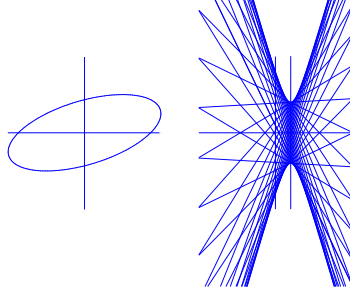


Figure 4: Ellipse(point-curve) $\rightarrow$ Hyperbola(line-curve).

as *pstars* and the sections with two unbounded branches as *hstars*, the mapping has analogous correspondences (see table 1). There are other point-line correspondences and dualities like *cusp* $\leftrightarrow$ *inflection point* shown in Fig. 6.

If you are still with us, congratulations. You are certifiably brave and perhaps incorrigibly masochistic. But we've only just begun!

Conic, <i>Gconic</i> $\rightarrow$	Image
ellipse, <i>estar</i>	hyperbola (Fig. 4), <i>hstar</i>
parabola, <i>pstar</i>	parabola, <i>pstar</i> or hyperbola, <i>hstar</i> with vertical asymptotes
hyperbola, <i>hstar</i>	ellipse (Fig. 5), <i>estar</i> , parabola, <i>pstar</i> , hyperbola, <i>hstar</i>

Table 1: Point curves, conics & *gconics* $\rightarrow$ Line curves, conics & *gconics*

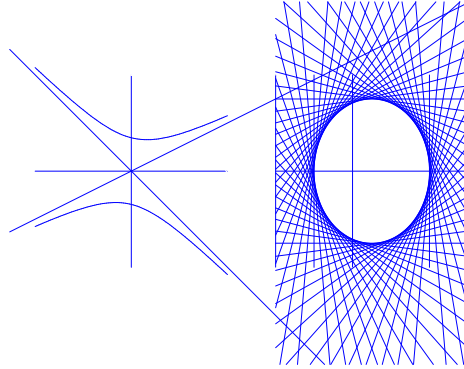


Figure 5: Hyperbola(point-curve) $\rightarrow$ Ellipse(line-curve), dual of Fig. 4 – Image depends on orientation of hyperbola.

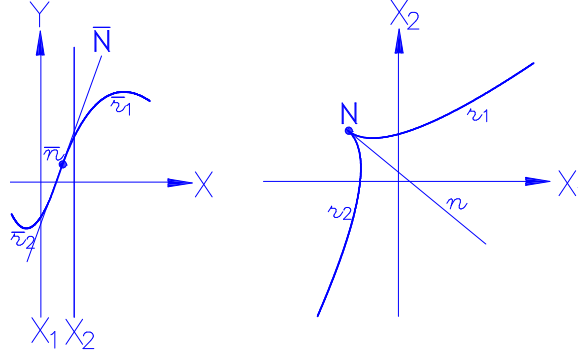


Figure 6: Cusp (point-curve) $\leftrightarrow$ Inflection point (line-curve) – correspondence is true in both directions.

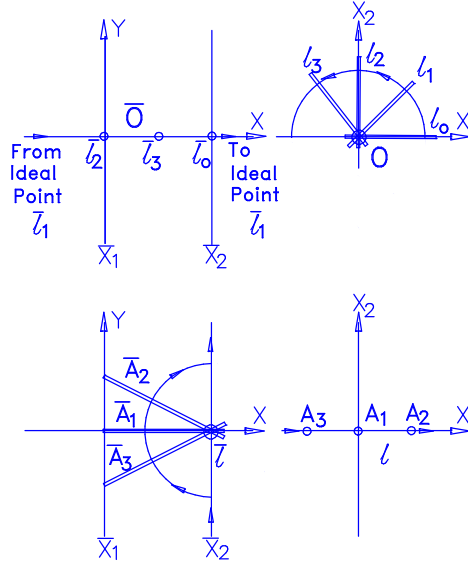


Figure 7: Duality between rotations and translations

2.1 Rotations and Translations under the Duality

Duality is also manifested in the transformations. With reference to Fig. 7, a line l originally in horizontal position, l_o , is rotated counterclockwise about one of its points which for convenience is taken as the origin O . Under the duality, the corresponding path of $\bar{l}$ is along the line $\bar{O}$ since in any of its rotated positions (say l_1, l_2, l_3 etc.) l still contains the point O . To better understand the corresponding motion of the point $\bar{l}$ we point out that in

the projective plane a line behaves like a closed curve “joined” at its ideal point. Hence due to the counterclockwise rotation, the point $\bar{l}$ moves in the direction of increasing x (i.e. to the right) until the line’s slope is 1 when $\bar{l}$ passes through the ideal point and “returns” to the Euclidean plane now to the left of the X_1 -axis but still moving to the right. Dually, a translation of point A in the positive direction along a line l (taken for convenience coincident with the x_1 -axis) corresponds to the rotation of the line $\bar{A}$ in the clockwise direction. There are also some useful higher dimensional analogues.

“Out of nothing I have created a strange new universe” Bolyai 1823

3 Linearity in R^N and the pesky indexing

In projective N-space, P^N , the general duality is *Point* $\leftrightarrow$ *Hyperplane*, ergo the *Point* $\leftrightarrow$ *Line* duality for $N=2$. For this reason the *Line* $\leftrightarrow$ *Point* correspondence of 2-D does not typically generalize into a *Line* $\leftrightarrow$ “*Something Nice*” relation. Parallel coordinates, in this respect, seem to have escaped Uncle Murphy’s mischief long enough to procreate some useful and intuitively satisfying generalizations to N-space.

3.1 Lines

Consider a line l in R^N described by

$$l_{i-1,i} : x_i = m_i x_{i-1} + b_i, \quad (6)$$

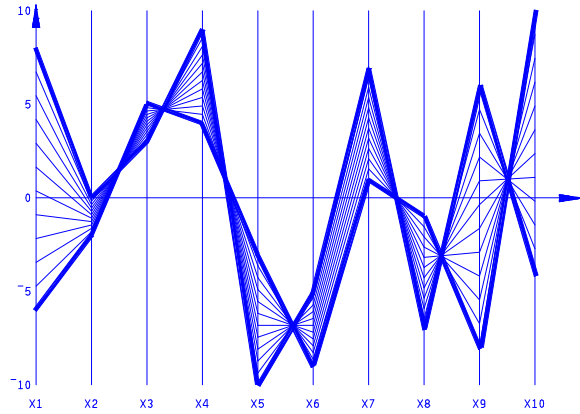


Figure 8: Line interval in 10-D. Heavier polygonal lines represent end-points.

for $i=2,\dots,N$. In the $x_{i-1}x_i$ -plane the relation labeled $l_{i-1,i}$ is a line and by the duality, Equation (5) with inter-axes distance of 1 unit, is represented in homogeneous coordinates by the point:

$$\bar{l}_{i-1,i} : ((i-2)(1-m_i) + 1, b_i, 1-m_i). \quad (7)$$

Hence there are $N-1$ such points, one for each of the linear equations in 6, which represent the N-D line l with the **indexing** being an essential part of the representation. This is not often appreciated and is probably responsible for the bulk of misunderstandings, arguments, untold mental distress which, at times, leads to divorce from ||-coords, of course, ... what did you think? Without the indexing, the points could represent any one of $(N-1)!$ lines corresponding to all possible ways the $(N-1)$ pairs of indices can be attached to the points. Specifically, $\bar{l}_{21}$ represents the linear relation between x_1 and x_2 for **this particular line** and so on. The number of indices is **two** as it pertains to a linear relation between two variables. Since display space is at a premium, the indexing which can often be deduced from the context, is not usually placed in the picture. In principle though it must *always* be accessible.

A polygonal line passing through all the $N-1$ points given by Equation (7) necessarily represents a point on the line l since the pair of y-coordinates a_{i-1}, a_i of its vertices simultaneously satisfy Equation (6) for $i=2,\dots,N$. Such is the case in Fig.8 where several polygonal lines representing points on an interval of a line in 10-D are shown. A number of construction algorithms, *crucially depending on the indexing*, based on this representation have been found including one for computing and **visualizing** the minimum distance between two lines. This representation was used to obtain Collision Avoidance Algorithms in conjunction with the new Air Traffic Control system in

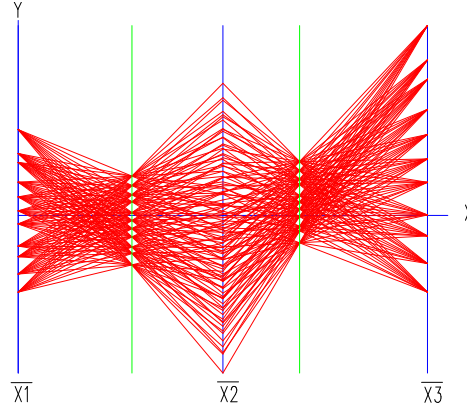


Figure 9: The polygonal lines correspond to points chosen from a regular grid on a plane π in 3-D. Notice the two vertical lines which correspond to a coordinate system on the plane π .

the USA (Patent # 4,823,272, # 5,058,024 and # 5,173,861) [18].

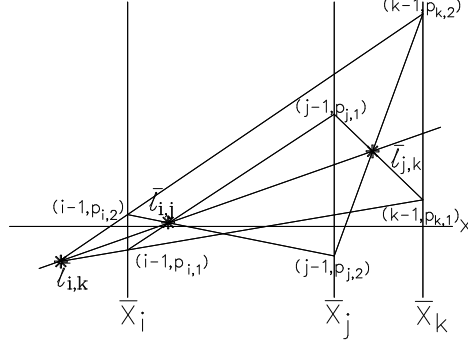


Figure 10: The collinearity for the 3 points $\bar{l}_{i,j}, \bar{l}_{j,k}, \bar{l}_{i,k}$. The two triangles, whose vertices are on the 3 parallel axes, are in perspective with respect to the ideal point in the vertical direction. The y-axis is offscale.

Al's 1st Meta-Murphy: Law of Attraction of Unfortunate Events
 "Once an unfortunate event occurs it attracts others of its kind."

3.2 Planes, hyperplanes and p-flats

It is in this section that most of the promised and eagerly awaited new applications are brought. They are designated with ** at the beginning of their description. In fact, this whole section is of considerable interest as an alternative way of doing Linear Regression.

In Fig. 9 we see a bunch of polygonal lines whose locus of intersections are two vertical lines. These represent *coplanar* points on a regular (does not need to be rectangular) grid on a plane. In fact, the vertical lines represent the grid's axes – shame, I just gave the proof away. Even in 3-D when looking at a set of points it is easy to miss the coplanarity unless one looks from the right viewpoint. In ||-coords the vertical lines pattern is unmistakable, does not depend on any viewpoint and works in N-space where the polygonal lines representing points on a hyperplane produce $N - 1$ vertical lines. This also gives rise to the interesting duality in R^3 :

rotation of a **plane** about a line $\leftrightarrow$ translation of a **point** on a line.

** This is a good time for a pause that refreshes. There is a promising application in computational statistics. There are implementations of the Grand Tour [28], [34], which also employ ||-coords. Apparently, the rotations are computed in orthogonal coordinates while the RESULTS are displayed in ||-coords. Yet if the problem is originally cast in ||-coords the computationally

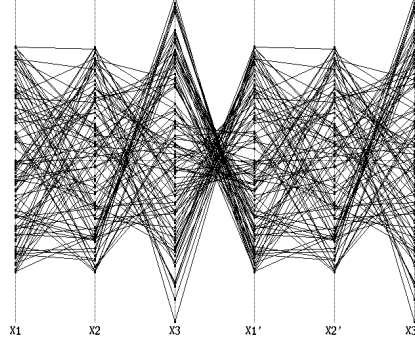


Figure 11: On the first 3 axes a set of polygonal lines representing a randomly sampled set of coplanar points in R^3 is shown.

costly rotations could be replaced, via this duality, by translations. This would remove or at least substantially reduce a serious computational barrier in the Grand Tour implementations (see also [3]).

Alas, points on a regular grid of a plane are not always available. How might coplanarity be recognized for a randomly chosen set of points on a plane? Eickemeyer [8] overcame this difficulty by describing a p -flat π^p – a plane of dimension $1 \leq p \leq N - 1$ – in terms of its $(p-1)$ -flats π^{p-1} (i.e. $\pi^{p-1} \subset \pi^p$) rather than directly in terms of its points. In fact, this gives rise to a *recursive*⁶ construction of p -flats where the recursion is on the dimension p . The construction algorithm depends crucially on a property of the points given by Equation (6), where for $i \neq j \neq k \in [1, 2, \dots, N]$ the three points

⁶To iterate is human, but to recurse is divine.

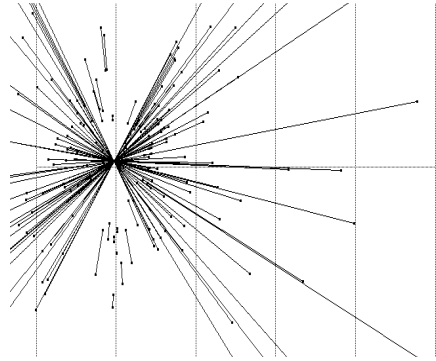


Figure 12: Coplanarity. From a collection of lines l in 3-D another collection of lines $\bar{L}$ is constructed from the $\bar{l}$'s using the 3pt collinearity property. The lines $\bar{L}$ intersect at a point $\iff$ the original lines l are coplanar.

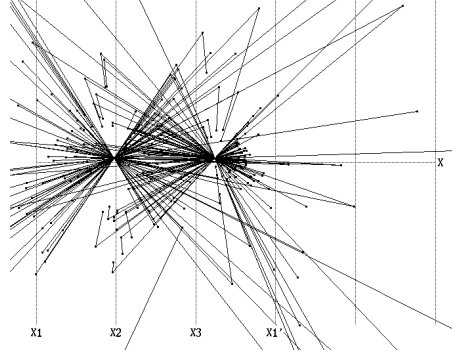


Figure 13: The plane π can be represented by two points.

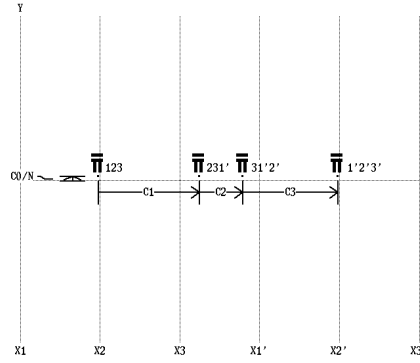


Figure 14: The distances between adjacent points are equal to the coefficients of $\pi : c_1x_1 + c_2x_2 + c_3x_3 = c_0$. The equation of the plane can be read from the picture!

$\bar{l}_{i,j}, \bar{l}_{j,k}, \bar{l}_{i,k}$ are always collinear. This can be seen by considering two points $P_r = (p_{1,r}, \dots, p_{N,r}) \in l$, $r = 1, 2$ on l and taking their projections on the x_i, x_j, x_k three-space as shown in Fig.10. The projected portions of the points form two triangles and the collinearity of $\bar{l}_{i,j}, \bar{l}_{j,k}, \bar{l}_{i,k}$ is obtained as a consequence of Desargue's Theorem (i.e. two triangles in perspective from a point are in perspective from a line). This property is essential for the construction of the representation of higher dimensional p-flats in N-space and will be referred as the *the 3-point collinearity property*.

To illustrate, consider a plane π^2 in R^3 given by:

$$\pi_2 : c_1x_1 + c_2x_2 + c_3x_3 = c_0. \quad (8)$$

From the representation of a line $l \subset \pi^2$ the points $\bar{l}_{1,2}, \bar{l}_{2,3}, \bar{l}_{1,3}$ are obtained, and by the 3-point collinearity property they yield a line $\bar{L}$. Taking another

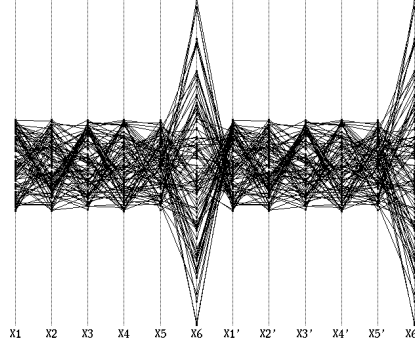


Figure 15: Polygonal lines on the $\bar{X}_1$ through $\bar{X}_6$ axes representing randomly selected points on a 5-flat $\pi^5 \subset R^6$

line $l' \subset \pi_2$ in the same way a corresponding line $\bar{L}'$ is obtained. The point:

$$\bar{\pi}_{123}^2 = \bar{L} \cap \bar{L}' = (c_2 + 2c_3, c_0, c_1 + c_2 + c_3) \quad (9)$$

has the remarkable property that for any other line $l'' \subset \pi_2$ the corresponding line $\bar{L}''$ also lies on $\bar{\pi}_{1,2,3}$.

As we will see, this property can also characterize coplanar points. Starting with a randomly chosen set $\mathcal{S}$ of points, whose representation is shown in the first 3 axes of Fig. 11, a set $\mathcal{L}$ of lines is constructed from pairs of points. Next the corresponding $\bar{L}$'s are constructed from the 3-point collinearity property providing the striking picture in Fig. 12. It turns out that the set of points $\mathcal{S}$ is coplanar $\iff$ all the $\bar{L}$'s intersect at a single point which is, in fact, the *envelope* of the $\bar{L}$ s. Basically, this picture identifies an infinite *family* of 2-flats intersecting at a line and containing the specific plane π^2 . To

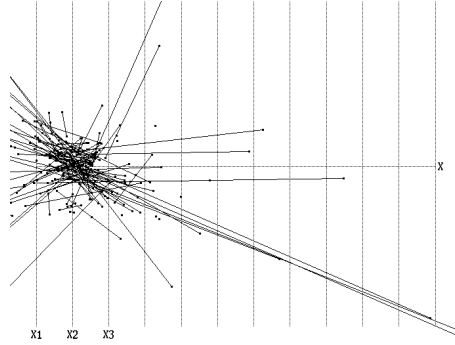


Figure 16: Portions $\bar{\pi}_{12}^{1i}, \bar{\pi}_{23}^{1i}$, of the 1-flats of π^5 constructed from the polygonal lines shown in Fig. 15. No pattern is evident.

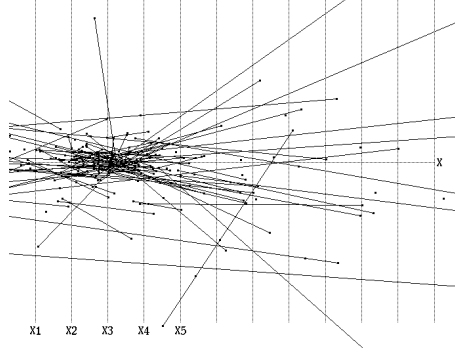


Figure 17: Portions, $\bar{\pi}_{1234}^{3i}$, $\bar{\pi}_{2345}^{3i}$, of the 3-flats of π^5 constructed from the polygonal lines obtained by joining $\bar{\pi}_{123}^{2i}$, $\bar{\pi}_{234}^{2i}$, $\bar{\pi}_{345}^{2i}$. Nothing yet ... but wait!

distinguish and characterize π^2 more information is needed. For this reason the axis $\bar{X}_1$ is translated to a new position $\bar{X}'_1$ one unit to the right of $\bar{X}_3$. This corresponds to rotating the original x_1 -axis to the equivalent x'_1 -axis position. Repeating the procedure now with the the polygonal lines on the $\bar{X}_2, \bar{X}_3$ and $\bar{X}'_1$ axes the point

$$\bar{\pi}_{231'}^2 = \bar{\ell}_{e_{12}} = \bar{\ell}_{e_{23}} = (3c_1 + c_2 + 2c_3, c_o, c_1 + c_2 + c_3). \quad (10)$$

is computed in the same way as the intersection of the $\bar{L}$'s.

These two points, eq. 9 and 10, completely identify π^2 . Note that there are now *3 indices* to distinguish these two points which represent a 2-flat in R^3 , from the points which represent a line and which have *2 indices*. The reason is that a line and π^2 are described by equations involving *two* and *three* variables respectively. To recoup, we started with points (which are 0-flats

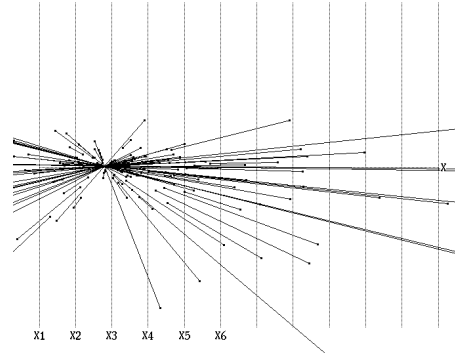


Figure 18: This is it! Portions, $\bar{\pi}_{12345}^{4i}$, $\bar{\pi}_{23456}^{4i}$, of the 4-flats of π^5 constructed from the polygonal lines joining $\bar{\pi}_{1234}^{3i}$, $\bar{\pi}_{2345}^{3i}$, $\bar{\pi}_{3456}^{3i}$, showing that the original points whose representation is shown in Fig. 15 are on a 5-flat in R^6 . The remaining points of the representation are similarly obtained.

i.e. 0-dimensional) on a 2-flat π^2 , then constructed lines (1-flats) on π_2 and from them obtained, via the envelopes of the $\bar{L}s$, of the the representation of the 2-flat still in terms of points. This outlines the recursive construction we alluded to earlier. In fact, the 3-point collinearity property generalizes to triples of points representing higher dimensional flats. The recipe for the recursion is the construction of the polygonal lines from the representation of the π^{q-1} and take their envelope to obtain the representation of the π^q until the procedure terminates.

Returning to the 2-flat π^2 we find that:

$$\bar{\pi}_{231'}^2 - \bar{\pi}_{123}^2 = 3(c_1, 0, 0). \quad (11)$$

with the constant being irrelevant due to the homogeneity of the coordinates. Continuing, the axes $\bar{X}_i$, $i = 2, 3$ are translated to the indicated positions and the additional points $\bar{\pi}_{31'2'}^2$, $\bar{\pi}_{1'2'3'}^2$ are computed in the same way resulting in the incredible property shown in Fig. 14 and eq. 11. All this works for R^N as the example in Fig. 15, Fig. 16, Fig. 17 and Fig. 18 shows for a 5-flat (hyperplane) $\pi^5 \subset R^6$.

Formally a p-flat is given by :

$$\pi^p : \pi_{i \dots (p+i)}^p : \sum_{k=i}^{p+i} c_{ik} x_k = c_{0i} \quad , \quad i = 1, 2, \dots, N-p. \quad (12)$$

This states that the p-flat π^p is described by N-p linearly independent equations, each denoted by $\pi_{i \dots (p+i)}^p$, which are the analogue of the description we used for lines in eq. 6 with the indexing being also a direct extension.

Theorem 3.3 : A p-flat in R^N described by the N-p linearly independent equations given by eq. (12) is represented by the $(N-p)p$ points :

$$\bar{\pi}_{i \dots N1' \dots (i-1)'}^p = \left(\sum_{k=1}^{p+1} d_{ik} c_{ik}, c_o, \sum_{k=1}^{p+1} c_{ik} \right), \quad (13)$$

where $i = 1, \dots, N-p$, the d_{ik} are the distances specified by $S_{1' \dots i'}$, that is the adjacent axes are 1 unit apart and the $\bar{X}_i'$ are placed successively also one unit apart for $i' = 1', \dots, p'$.

Note the important special cases:

- when $p = 1$ (i.e. for lines) Equation (13) yields $N - 1$ points having *two* indices,
- when $p = 2$ and $N = 3$ (i.e. plane in 3-D) Equation (13) yields *two* points having *three* indices, and
- when $p = N - 1$ (i.e. hyperplane) Equation (13) yields $N - 1$ points having N indices.

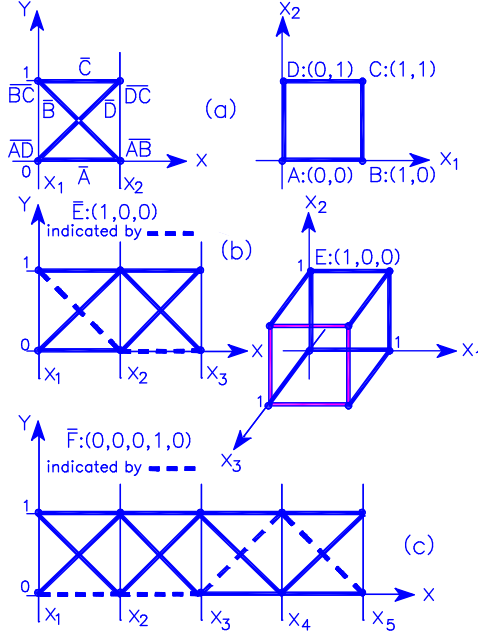


Figure 19: (a) A square, (b) 3-D cube and (c) 5-D hypercube all with unit side

** The indexed point representation of p -flats turns out to be robust in the sense that it still works for “approximate” coplanarity. More precisely, when points are nearly coplanar (i.e. perturbing the coefficients of the planar equation to generate a wavy “slab”), the construction in Fig. 12 yields lines $\bar{L}$ intersecting at several but *closely* spaced points and the pattern is unmistakable. That is, this construction can function as a “slab” detector. It was recently found [22] that the points representing all possible planes “fitting” a given set of points form a convex hexagon whose vertices provide tight error estimates for the deviation (including directions) from coplanarity. The generalization for N -space is that the points representing p -flats lie in a convex $2N$ -agon whose vertices provide the extreme values of the coefficients. In fact, the construction can be modified to identify or cluster points in several “slabs”. This seems like a fruitful area for investigation to improve the algorithm’s numerical efficiency. Gennings et al [10] in an elegant application used this property to detect the interaction of drugs in chemotherapy.

** It is time to go out on a limb with a claim supported by the above. **Low dimensional earthlings can not, in general, visualize p -dimensional objects directly from their randomly chosen points.** Rather, starting from the points which are 0-dimensional, $p-1$ dimensional *components* are assembled *recursively* from which the object can be visualized. As we saw above this process “works” at least for p -flats. Of course, this is not a *proof* that a more direct method of visualization does not exist. The caveat “in

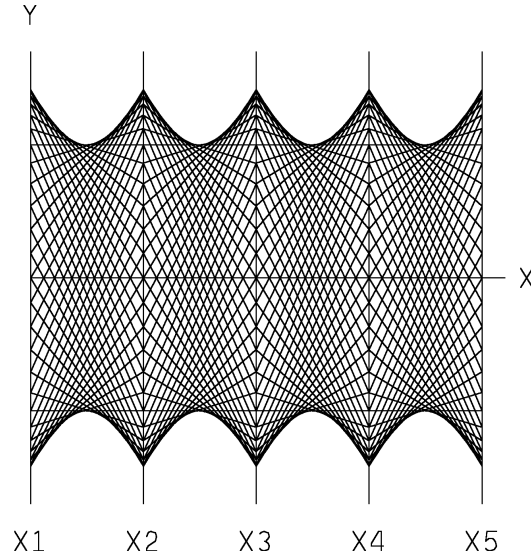


Figure 20: A Sphere in 5-D

general” is meaningful in the sense that if the sampled points have special properties, as for example in Fig. 9, the object may be “visualizable” directly from collections of such points. The next section builds on this approach.

AI’s 2nd Meta-Murphy: Law of Inopportune Timing

“If a fortunate event occurs at all it is likely to come a bit too soon or a bit too late.”

4 Polytopes, Hypersurfaces and Applications

From the representation of p -flats a number of construction algorithms are obtained including testing whether a point is on a given p -flat (if not on what side it is), intersections as well as an extension of the duality between rotations about a line and translations of a point along a line. Simple polytopes can also be easily constructed and displayed as shown in in Fig.19. This figure actually shows not only the vertices but also the edges and faces as well as the cube’s interior. However, for this to be accomplished the complete indexing of the hypercube’s elements must be utilized – it is fun, try it!

******Recently Chatterjee [4] methodically studied the transformation dualities mentioned earlier and showed that a *projective transformation* (i.e. translation, rotation, scaling and perspective) becomes another projective transformation. So a multidimensional object can still be recognized as the *same* object after it has been acted on by such a transformation. Further, using Eickemeyer’s representation, he obtained representations for polytopes

in N-D where adjacencies of the faces of various orders can be directly visualized. He extended these to obtain some very useful notions for proximity and tolerances having applications in computer vision and solid modeling.

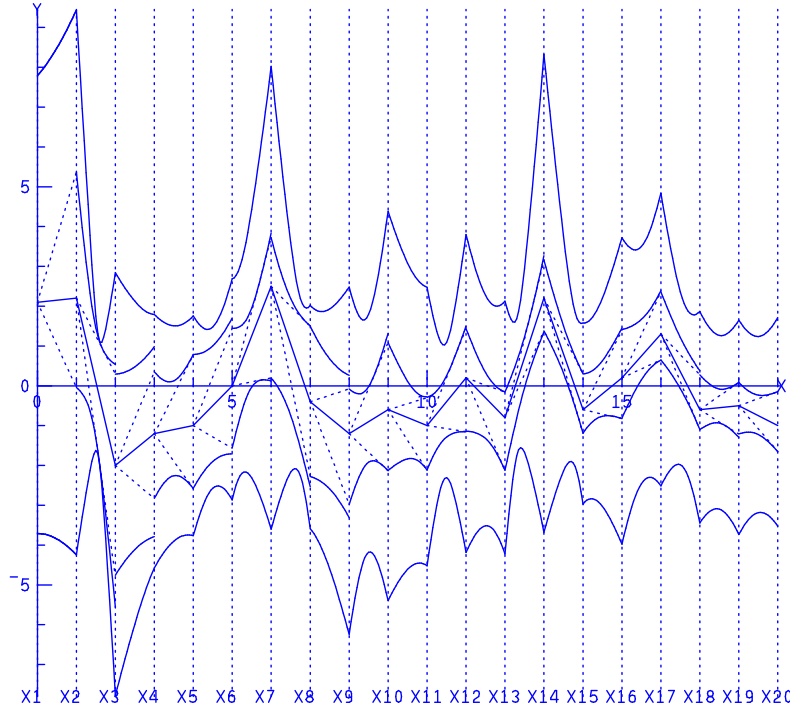


Figure 21: A Convex Hypersurface in 20-D and Interior Point Algorithm.

4.1 Hypersurfaces – “Ambiguous Representation”

Originally the representation of smooth hypersurfaces was obtained as the envelope of the polygonal lines representing the surface points. We refer to this as the *ambiguous representation* since the the same 2-D pattern can be obtained in this way from a class of somewhat similar hypersurfaces. For example, the pattern shown in Fig. 20 is the envelope obtained in this way from the points of a sphere in 5-D. The *same pattern* also results from the surface formed by the intersection of 5 cylinders all having the same radius and all axes intersecting at the point corresponding to the sphere’s center. The true representation of hypersurfaces requires recursion on the dimensionality, as alluded to earlier, and is described in the last section – at last, it is about time! There exists a very useful algorithm for constructing and display-

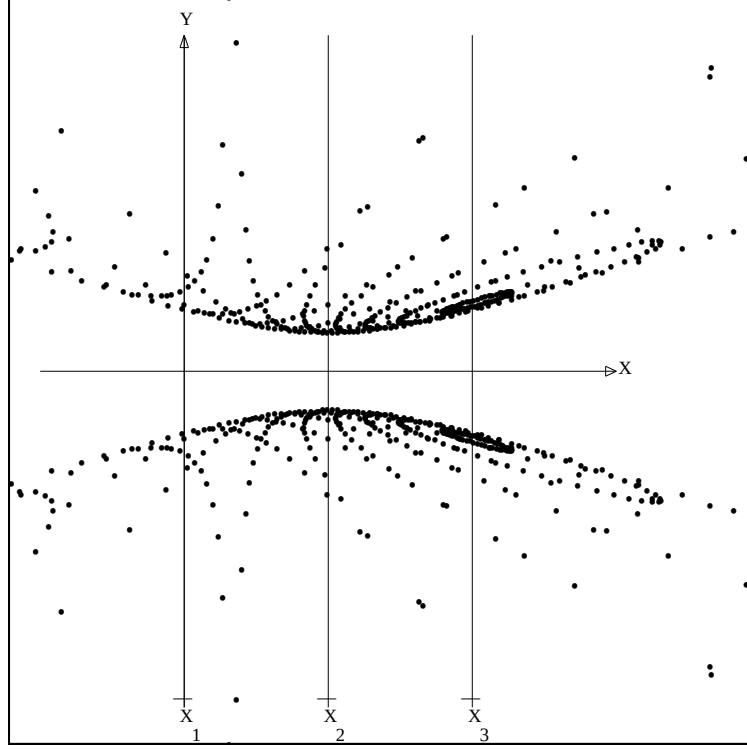


Figure 22: A Sphere in 3-D represented by its tangent planes (points). The hyperbolic pattern of the envelopes indicates that the object is **convex**.

ing *interior points* based on the naive representation of hypersurfaces which are piecewise convex. It is illustrated here with reference to Fig.21 where the naive representation of a convex hypersurface $\mathcal{S}$ in 20-D is shown. The polygonal line $\bar{P}$ representing a point $P \in \mathcal{S}$ is constructed by choosing the coordinates of P *sequentially* starting with the first one (which in principle can be anyone of the N coordinates). On the $\bar{X}_1$ -axis the range I_1 of x_1 **for all the points** in $\mathcal{S}$ is displayed. A value $x_1 = \alpha_1$ in this range is chosen, as shown, and the algorithm draws tangents to the upper and lower envelopes of $\bar{\mathcal{S}}$ between the $\bar{X}_1$ and $\bar{X}_2$ -axes extended where necessary. By fixing the value of $x_1 = \alpha_1$ the dimensionality of the resulting hypersurface, say $\mathcal{S}_1$, has been reduced to 19 and the corresponding envelopes of $\bar{\mathcal{S}}_1$ are computed and displayed as shown between the $\bar{X}_2$ and $\bar{X}_3$ -axes. This necessitates that the precise description for $\mathcal{S}$ be available *internally* for the algorithm, hence compensating for the ambiguity in the representation $\bar{\mathcal{S}}$. Further, the algorithm

needs to evaluate the resulting envelopes at each stage. On the $\bar{X}_2$ -axis the *available range*, I_2 , of x_2 **for all** points of $\mathcal{S}_1$ is displayed. A value $x_2 = \alpha_2$ is chosen and the process is repeated for the resulting 18-dimensional hypersurface $\mathcal{S}_2$ and so on. At the i th-step the resulting range I_i has “inherited” all the previously imposed constraints and the construction is repeated by choosing $x_i = \alpha_i \in I_i$ with the algorithm then supplying the tangents to the envelopes of $\bar{\mathcal{S}}_i$ between the $\bar{X}_i$ and $\bar{X}_{i-1}$ -axes, as well the envelope of $\bar{\mathcal{S}}_{i+1}$. The algorithm ends when the construction of an *interior* point P is completed for $i = D$ the dimensionality of the hypersurface. For an *exterior* point $\bar{P}$ must intersect the intermediate envelopes $\bar{\mathcal{S}}_i$ for an i , while P is a boundary point if $\bar{P}$ is tangent to the intermediate envelope for an i .

Hung [11] recently obtained the representation of smooth surfaces by considering them as the envelope of their tangent hyperplanes. That is, in R^N a hyperplane is represented by $N - 1$ points with N indices all being adjacent permutations of $[1, 2, \dots, N]$. So from each point of a smooth hypersurface σ , $N - 1$ points are obtained for the representation of the tangent hyperplane at that point. The representation $\bar{\sigma}$ consists of N point sets, $\bar{\sigma}_i$, where the i stands for one of the permutations. On these, certain grids can be constructed for the reconstruction of the points of σ . It turns out that the $\bar{\sigma}_i$ are *curves* (i.e planar sets with just a boundary and no interior points) $\iff \sigma$ is a *developable* surface. Further a grid on the $\bar{\sigma}_i$ can be constructed with one family of grid curves being *straight lines* $\iff$ the surface is *ruled*.

Based on this representation Chatterjee [4] generalized the results on gconics (see section on duality in 2-D) so that, roughly speaking, a body in R^N is convex if the envelope of the points representing its faces of all orders forms an *hstar*. For the first time, convexity in N-D can be seen! For an example see the representation of a sphere in 3-D shown in Fig. 22.

Hypersurfaces can serve as geometrical models for relations among several variables, and with this representation they provide a very useful way for modeling and visualizing complex processes. With the interior point algorithm applied interactively it has been possible to explore sensitivities, interdependencies, the effect of constraint imposition *downstream* (i.e. on the remaining variables), criticalities of various kinds and others. Developing and implementing this capability in suitable software is likely to bring a paradigm-shift in the *discovery and visualization of multivariate relations* among large data sets. A simplified model for the economy of a real country was obtained in this way [15] and easily provided interesting insights.

Finally, since I am often asked, the most fruitful research area on parallel coordinates is on the representation of *space* curves and a wider class of hypersurfaces. The ability to visualize multivariate **relations** is what particularly distinguishes parallel coordinates from other multidimensional visualization methodologies and its development will help bring us closer to envision ...
Nvision©— (all rights reserved).

Epilogue

I am very grateful to the referees for many constructive remarks, suggestions, corrections and improvements. Their most justified request was to replace the abstract by a serious and more informative one as you can see for yourself.

The Original Abstract

This is a light opera about **Parallel Coordinates**. It opens with an emulation of the ancient witch-doctors' rituals, exorcising, cursing, pursuing, projecting, scatterploting and mutilating (i.e. "reducing") *Dimensionality*. From its lofty palace Dimensionality (*big-D*), bemused with this frenzy and munching on PLS, PCA and sushi by Taguchi, hurls volleys of new *dimensionalitis* strains earthwards to cause great havoc as in times past. But prehistory does not repeat itself. Disguised as asparagus spears, parallel coordinates appear, **one for each dimension**, and render harmless the dimensionalitis viruses by transforming them **polygonally**. From the polygonal lines the **envelopes** then fashion intricate patterns **representing relations**. Big-D, whose ire by now is aroused, sends higher and even hyper-dimensions (*h-D's*) to overwhelm the low-dimensional earthlings. More parallel coordinates unfold and **recursively** – (don't confuse with re-*CURSE*) – tame the h-D's; embedding them in the **Projective** plane and shackling them with **indices**. **Duality** triumphs, the asparagus spears do a coordinated ballet in parallel, the choir sings the rousing ecumenical blessing for Dimensionality, the curtains fall and everybody lives happily ever after ... well almost everybody.

"Better is the end of a thing than the beginning of it" Ecclesiastes 7:8

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